

Flocking Control of Multiple AUVs Based on Fuzzy Potential Functions

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Abstract— This paper presents the formulation of a flocking control algorithm for a group of autonomous underwater vehicles (AUVs). A leader-follower control strategy is employed to flock a group of AUVs along a predefined desired path. In this approach, leader-AUVs are assumed to have global knowledge of the desired trajectory and the follower AUVs are not provided with this information. For keeping all the AUVs connected in a group, a flocking center is estimated. This flocking center is a virtual point whose position at any instant of time can be predicted by using a consensus algorithm. The controllers for the leader and follower AUVs are developed by implementing mathematical and fuzzy artificial potential functions. These potential functions are the inter AUV distance dependent variables. Fuzzy artificial potential functions are calculated by using Mamdani's fuzzy logic technique. A group of four AUVs is considered for analyzing the efficacy of the developed control algorithm. Simulations are carried out both in obstacle-free and obstacle-rich environments. From the obtained results, it is observed that the proposed fuzzy flocking control algorithm provides effective co-operative motion control of multiple AUVs along the desired paths and avoid obstacles successively. It is also observed that the flocking controller developed based on fuzzy artificial potential function outperforms the controller with mathematical potential functions despite uncertainties owing to external disturbances.

Index Terms— Consensus Algorithm, Cooperative Control, Artificial Potential Function, Flocking Algorithm, Graph Theory, Mathematical Potential Function, Fuzzy Potential Function.

NOMENCLATURE

$\{B\}$ Body fixed frame of reference
 $\{I\}$ Inertial frame of reference
 ζ_i Position and orientation vector of the i^{th} AUV
 v_i Velocity vector
 u_i, v_i, w_i Linear velocity vectors
 p_i, q_i, r_i Angular velocity vectors
 X_i, Y_i, Z_i Force vector
 K_i, M_i, N_i Vector of moments

M_i Inertia matrix including added mass
 $C_i(v_i)$ Matrix of Coriolis and centripetal terms
 $D_i(v_i)$ Hydrodynamic damping and lift matrix
 $g_i(\zeta_i)$ Vector of gravitational forces and moments
 F_i Vector of forces
 F_i^d Vector of oceanic disturbances
 $J_i(\zeta_i)$ Velocity transformation matrix
 r_{ij} Distance between the i^{th} and the j^{th} AUVs
 d Desired distance between the i^{th} and the j^{th} AUVs
 n_l Number of leader AUVs
 n_f Number of follower AUVs
 ζ_i^c Flocking center of the i^{th} AUV
 ζ_c Average of position coordinates of AUVs
 ζ^c Vector of flocking centers
 ε Error between ζ^c and average position coordinate of all AUVs
 $\mathcal{L}$ Lyapunov function
 $U(\zeta)$ Potential function
 $U_{i,att}^c(\zeta_i, \zeta_i^c)$ Attractive potential function between the i^{th} AUV and its flocking centers
 $F_{i,att}^c$ Force of attraction between the position of the i^{th} AUV and position of corresponding flocking center
 $F_{i,att}^d$ Force of attraction between the position of the i^{th} AUV and position of desired path
 F Total potential function of the whole system
 ζ_i^l Position and orientation vector of the i^{th} leader AUV
 ζ_i^f Position and orientation vector of the i^{th} follower AUV
 ζ_i^{lc} Flocking center of the i^{th} leader AUV
 ζ_i^{fc} Flocking center of the i^{th} follower AUV

I. INTRODUCTION

RESearch on cooperative control of AUVs is an important

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This work was supported by Naval Research Board, Defense Research Development Organization (DRDO), Government of India (No. DNRD/05/4003/NRB/160).

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and popular topic owing to several control challenges due to nonlinear dynamics, complex structure of AUVs and their pertinent applications. The major applications of AUVs can be found in commercial, military and research fields. The AUVs are useful in oil and gas industries, survey of ocean floor, post-lay pipe surveys, inspection of mines and to check the safety of the needy areas. Apart from these, AUVs are also deployed in anti-submarine warfare and used for detection of manned submarines [1]. It is difficult to carry out the complex underwater missions by a single AUV. Hence a group of multiple AUVs is deployed to solve these difficult tasks easily, which are beyond the scope of a single AUV. This method of working on a group of multiple AUVs is known as flocking of AUVs. Some of the typical methods of flocking control reported in the literature are presented here. Distributed flocking control of multiple agents with the obstacle avoidance is deliberated in [3] and [27]. The flocking control proposed in [3] is modified in [34] where a fraction of uniform agents is used to drive multiple agents. Here, the velocities of the agents are not constants but considered to be the same as the desired velocities. The virtual leader concept is employed for flocking control of multiple agents in [3], [16], [17], [20], [21], [25], [29], and virtual force with pseudo-leader concept is proposed in [30]. Flocking algorithms for a group of multiple agents is accomplished using numerous virtual leaders in [7]. Here the whole group is divided into many subgroups with virtual leaders and these subgroups have velocities equal to that of the virtual leaders. A group of non-holonomic car like vehicles are used in the development of flocking algorithm in a constrained environment in [9] where the centralized motion planner is employed for maneuvering of the flockmates. Assuming that every agent accepts a certain number of flockmates, a flocking algorithm for multiple agents has been developed in [10]. Based on this algorithm, all agents are assumed to be connected and the information exchange between flockmates takes place through a connected graph topology. The potential function based flocking algorithm for multiple mobile agents and for AUVs have also been developed using a fuzzy algorithm in [13], [33], and [5] respectively. Controllers based upon neural-fuzzy approach is proposed in [32] and the controllers are developed using attractive/repulsive potential functions are presented in [18], [24], [26], and [36]. Leader-follower flocking algorithm based on the potential function approach is developed in [12]. Decentralized flocking control algorithms for both fixed and switching network topologies are described in [6]. A decentralized flocking controller for operating a Cucker-Smale multi-agent system is designed in [8] using model predictive scheme and input constraints. A type-2 fuzzy logic technique is used for the architecture of flocking structure of multiple autonomous agents with considering noisy sensor measurement is presented in [35]. In [31], a distributed flocking algorithm for a group of multiple agents is developed using different behaviors of the agents present in the group. Flocking control plays an important role in the field of biomedical engineering. Flocking of multiple micro-particles using velocity saturation method in proposed in [28]. In most of the cases, the flocking control of multiple agents is carried out

in infinite space which has lack of practical applicability. But the flocking of multiple mobile robots was conducted in a bounded space in [4], where the practical implementation is possible.

From the literature review, it is observed that, owing to the applications of the AUVs in aforesaid important fields, research on AUVs is very challenging. Further, it is advantageous to deploy a group of multiple AUVs i.e. using of flocking of AUVs to perform a complex task which is beyond the scope of a single AUV. The advantages of flocking of multiple AUVs over a single AUV includes; increase of robustness and efficiency of the system while providing redundancy, reduction of the AUV system cost, reconfiguration ability and structural flexibility of the system. Also it is observed that, flocking of mini robots, mobile agents and car like vehicles are reported in a number of references. Although the flocking algorithm using consensus method and leader-follower approaches for the navigation of multiple autonomous agents is a very well-studied topic in the literature of robotics and control, the application of this algorithm in the field of underwater vehicle is the rarest one. However a few works on flocking control of multiple AUVs is reported in literature. No example of an experimental study of the flocking control of a group of AUVs is found in literature. These advantages of flocking of AUVs as well as the lack of literature on this motivate the authors for the development of a flocking control algorithm for a group of AUVs. Based on these motivations, the flocking control algorithms are developed and are presented here. Hence the contribution of this paper lies in the development of leader-follower flocking algorithm for a group of multiple AUVs using artificial potential function methods. Two types of artificial potential functions are considered here; mathematical potential functions and fuzzy potential functions.

The innovation of this paper lies on the application of fuzzy logic together with fuzzy potential functions to design efficient controller that would generate required amount of thruster force to steer the AUVs to track the desired trajectory. The superiority of the proposed fuzzy potential based controller over the mathematical potential function based controller is also demonstrated. Owing to the new ness of this work in proposing fuzzy potential based controller for AUV applications which will add value to the existing literature that has least found on the said subject area.

When formulating the control law it is considered that, the individual flockmate stays connected with the group through the flocking center whose position is calculated by using a consensus algorithm. The flocking control law is developed by using artificial potential functions. This controller is able to drive flocking structure to move along the desired path without facing any collision among them. In this paper there are two types of artificial potential functions are considered; mathematical potential functions and fuzzy potential functions. Simulations are carried out for both the flocking controllers both in obstacles-free and obstacles-rich environment. The performances of both the controllers based on flocking approach time and safe distance between AUVs for collision avoidance are compared.

The rest of the paper is organized as follows. The modelling of an AUV is presented in Section II and the problem formulation is described in Section III. In Section IV, mutual potential function analysis with the flocking algorithm for multiple AUVs is developed. Analysis of mathematical and fuzzy separation potential functions are presented in Section V. Numerical simulations are reported in Section VI to show the efficacy of the developed controllers. Conclusions are presented in Section VII.

II. KINEMATIC AND DYNAMIC MODELLING OF AN AUV

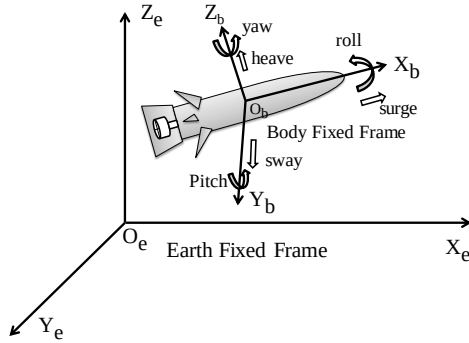


Fig. 1. Schematic diagram of an AUV showing forces and torques

Kinematics and dynamics of an AUV in six degrees of freedom (DOF) i.e. the motion of AUV in three dimensional spaces can be expressed as given in (1). Usually two types of reference frames, i.e. a body fixed frame of reference {B} and an earth fixed frame of reference or inertial frame of reference {I} are considered as shown in Fig. 1. X_e, Y_e, Z_e in the Fig.1 are the X, Y, and Z-axis components of the earth fixed frame of reference.

Let there are n number of AUVs are considered in the flocking group. The motion of the i^{th} AUV ($i \in \{1, 2, 3, \dots, n\}$) in six DOF can be described by using the following vectors.

$$\begin{aligned} \zeta_i &= [x_i, y_i, z_i, \phi_i, \theta_i, \psi_i]^T \\ v_i &= [u_i, v_i, w_i, p_i, q_i, r_i]^T \\ F_i &= [X_i, Y_i, Z_i, K_i, M_i, N_i]^T \end{aligned} \quad (1)$$

where ζ_i : position and orientation vector of i^{th} AUV, x_i, y_i, z_i : coordinates of position, ϕ_i, θ_i, ψ_i : orientation along longitudinal, transversal and vertical axes respectively, v_i : velocity vector with coordinates in the body-fixed frame. u_i, v_i, w_i : linear velocities, p_i, q_i, r_i : angular velocities. X_i, Y_i, Z_i : forces, K_i, M_i, N_i : angular moments.

The dynamics and kinematic equations of motion can be expressed as [14],

$$\dot{\zeta}_i = J_i(\zeta_i) v_i \quad (2(a))$$

$$M_i \dot{v}_i + C_i(v_i) \dot{v}_i + D_i(v_i) \dot{v}_i + g_i(\zeta_i) = F_i \quad (2(b))$$

In presence of external perturbations i.e. oceanic disturbances:

$$M_i \dot{v}_i + C_i(v_i) \dot{v}_i + D_i(v_i) \dot{v}_i + g_i(\zeta_i) = F_i + F_i^d \quad (2(c))$$

where M_i : inertia matrix including added mass, $C_i(v_i)$: matrix of Coriolis and centripetal matrix including added mass, $D_i(v_i)$: hydrodynamic damping and lift matrix, $g_i(\zeta_i)$: vector of gravitational forces and moments, F_i : vector of forces and moments acting on the i^{th} AUV in the body-fixed frame. F_i^d : vector of oceanic disturbances. Define the transformation matrix $J_i(\zeta_i)$ that transforms earth fixed frame to AUV frame as

$$J_i(\zeta_i) = \begin{bmatrix} J_1(\zeta_i) & 0_{3 \times 3} \\ 0_{3 \times 3} & J_2(\zeta_i) \end{bmatrix} \quad (3)$$

where
where

$$J_1(\zeta_i) = \begin{bmatrix} c\psi_i c\theta_i & -s\psi_i c\theta_i & c\psi_i s\theta_i s\phi_i & s\psi_i s\theta_i s\phi_i & c\psi_i c\phi_i s\theta_i \\ s\psi_i c\theta_i & c\psi_i c\theta_i & s\phi_i s\theta_i c\psi_i & -c\psi_i s\theta_i c\psi_i & s\theta_i s\psi_i c\phi_i \\ -s\theta_i & c\theta_i s\phi_i & c\theta_i c\phi_i & & \end{bmatrix} \quad (4)$$

$$\text{and } J_2(\zeta_i) = \begin{bmatrix} 1 & s\phi_i t\theta_i & c\phi_i t\theta_i \\ 0 & c\phi_i & -s\phi_i \\ 0 & s\phi_i / c\theta_i & c\phi_i / c\theta_i \end{bmatrix} \quad (5)$$

with $s(\bullet) = \sin(\bullet)$, $c(\bullet) = \cos(\bullet)$, $t(\bullet) = \tan(\bullet)$.

III. PROBLEM FORMULATION

Consider there are four AUVs take parts in cooperative motion as shown in Fig. 2.

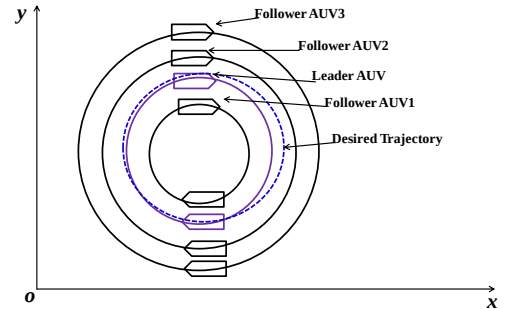


Fig. 2. Schematic presentation of flocking of four AUVs

The objective of this paper is to develop a distributed flocking control algorithm for each AUV in the group such that the multi-AUV system navigates to track the desired trajectory given by

$$\zeta_d = [x_d, y_d, z_d, \phi_d, \theta_d, \psi_d]^T \quad (6)$$

where ζ_d : desired linear position and orientation vector, x_d, y_d, z_d : coordinates of the desired position, ϕ_d, θ_d, ψ_d : desired orientations.

Trajectory tracking starts from an arbitrary initial connected topological configuration which is presented as;

$$\zeta_{i0} = [x_{i0}, y_{i0}, z_{i0}, \phi_{i0}, \theta_{i0}, \psi_{i0}]^T \quad (7)$$

where ζ_{i0} : initial position and orientation vector of i^{th} AUV, x_{i0}, y_{i0}, z_{i0} : coordinates of the initial position of i^{th} AUV, $\phi_{i0}, \theta_{i0}, \psi_{i0}$: initial orientations.

The positions of the AUVs should asymptotically converge

to the desired trajectory i.e. $\lim_{t \rightarrow \infty} \zeta_i \rightarrow \zeta_d$ and to maintain a certain distance between neighbors. It is intended to avoid collision among the AUVs with their flockmates during maneuvering i.e.

$$\lim_{t \rightarrow \infty} r_{ij} = d, \text{ where } r_{ij} = \|\zeta_i - \zeta_j\| \quad (8)$$

where $\|\bullet\|$: Euclidian norm, r_{ij} : distance between i^{th} and its neighbor j^{th} AUVs, d : desired distance to be maintained between the i^{th} and the j^{th} AUVs, t : time. The number of neighbor AUVs of i^{th} AUV is n_{ni} such that $j \in \{1, 2, 3, \dots, n_{ni}\}$. The distance between the i^{th} and the j^{th} AUVs is always greater than a minimum distance to ensure the avoidance of collision between the flockmates. Similarly the distance between the i^{th} and the j^{th} AUVs is always less than a maximum distance so that the flockmates will not diverge and stay within the group.

IV. DEVELOPMENT OF FLOCKING CONTROL OF AUVs

A. Graph theory

The degree matrix $D(G)$ of a graph G is the diagonal matrix $\{\text{dig}(ii)\}$ which has the information about each vertex. The Laplacian matrix $L(G)$ may be defined as [15],

$$L = L(G) = D(G) - A(G). \quad (9)$$

The Laplacian matrix possesses many topological properties and is a semi-definite matrix. The algebraic multiplicity of its eigenvalue is equal to the number of connected components in the graph. The n -dimensional eigenvector associated with the zero eigenvalue is the vector of ones. Hence $L(G)$ is always positive semi-definite and symmetric for undirected graph G and satisfies the quadratic form $b^T L b$ which is given by

$$b^T L b = \frac{1}{2} \sum_{i=1}^n \sum_{j \in C_i} a_{ij} (b_j - b_i)^2 \quad (10)$$

where $b = (b_1, b_2, \dots, b_n)$ for connected undirected graph G , the Laplacian has a single eigenvalue and the corresponding eigenvector has the dimension of $[1, 1, \dots, 1]^T$.

B. Consensus Theorem

Consider there are n AUVs in a group. Out of these n_l numbers of AUVs are considered as leaders and n_f number of AUVs as followers. Hence, $n = n_l + n_f$. Rather than the follower AUVs following the leader AUV directly, the positions of all vehicles in the group directed towards a virtual point called the flocking center. Each AUV sends its present state information to all the neighbor AUVs. The idea behind the flocking algorithm as follows. All AUVs will remain connected within the flocking structure during motion. It is possible through connection of each AUV with their neighbor AUV via the flocking center. The position of the flocking center is an approximation and can be calculated by taking the average of the position of all AUVs in the group. All AUVs obtain the information about the position of the flocking center through

communication topology established among neighbor AUVs. Hence all the AUVs are able to remain connected with the flockmates within the group. A consensus algorithm is presented here to keep neighbor AUVs connected within the group.

By following [2], [12], [19], [22], [23], and (10), the appropriate consensus algorithms developed to estimate the position of the flocking center ζ_i^c of the i^{th} AUV as

$$\dot{\zeta}_i^c = \sum_{j=1}^{n_{ni}} (\zeta_i^c - \zeta_j^c) + (\zeta_j^c - \zeta_i^c) \quad (11)$$

where ζ_j^c is the estimated position of the flocking center of the j^{th} AUV. In vector form this can be presented as

$$\dot{\zeta}^c = \zeta - (L+1)\zeta^c \quad (12)$$

where $\zeta^c = [\zeta_1^c, \zeta_2^c, \dots, \zeta_n^c]^T$ is a matrix formed by the vectors of flocking centers of all the individual AUVs, $\zeta = [\zeta_1, \zeta_2, \dots, \zeta_n]^T$ is a matrix formed by the vectors of the positions of all the individual AUVs.

The whole flocking group is to follow the desired trajectory i.e. the average center of the group will follow the desired trajectory. The average position of coordinates (ζ_c) of all AUVs is defined as

$$\zeta_c = \frac{1}{n} \sum_{i=1}^n \zeta_i. \quad (13)$$

To follow the desired trajectory by AUVs, ζ^c should asymptotically converge to ζ_c and $\zeta_i^c \rightarrow \zeta_i$.

For global asymptotic stability, the average of the coordinates of positions of AUVs (ζ_{av}) is presented by

$$\zeta_{av} = \frac{1}{n} O_{n \times n} \zeta \quad (14)$$

where $O_{n \times n}$ is the matrix of ones, $O_{n \times n} \in R^{n \times n}$.

Hence the error between ζ_c and average coordinate is presented by

$$\varepsilon = \zeta_{av} - \zeta^c. \quad (15)$$

Therefore

$$\dot{\varepsilon} = \dot{\zeta}_{av} - \dot{\zeta}^c. \quad (16)$$

$$\text{where } \dot{\zeta}_{av} = \frac{1}{n} O_{n \times n} \dot{\zeta}$$

Substituting the value of $\dot{\zeta}^c$ from (12) in (16), it gives

$$\dot{\varepsilon} = \dot{\zeta}_{av} + L\zeta^c + \zeta^c - \zeta. \quad (17)$$

From (18), $\zeta^c = \frac{1}{n} O_{n \times n} \zeta - \varepsilon$. Using the value of ζ^c in (17)

and solving, it gives

$$\dot{\varepsilon} = \frac{1}{n} O_{n \times n} (\dot{\zeta} + \dot{\zeta}) - L\zeta_{av} - \varepsilon(L+I) - \dot{\zeta} \quad (18)$$

where I is the identity matrix.

But in a connected graph [15],

$$L\zeta_{av} = 0. \quad (19)$$

Hence (21) becomes

$$\dot{\varepsilon} = \dot{\zeta}_{av} - \varepsilon(L+I) - \zeta + \zeta_{av}. \quad (20)$$

The k^{th} element of the vector $(\dot{\zeta}_{av} + \zeta_{av} - \zeta)$ is given by

$$\dot{\zeta}_{k_{av}} - \frac{1}{n} \sum_{k=1}^n (\zeta_k - \zeta).$$

where $\dot{\zeta}_{k_{av}} = \frac{1}{n} \sum_{k=1}^n \dot{\zeta}_k$. Further it is assumed that, the

position and velocity vectors of AUVs are uniformly bounded. Hence it can be shown that

$$\left\| \dot{\zeta}_{k_{av}} - \frac{1}{n} \sum_{k=1}^n (\zeta_k - \zeta) \right\| \leq \gamma \quad (21)$$

where γ is positive definite constant.

Consider a Lyapunov candidate function as $\mathcal{L} = \frac{1}{2} \varepsilon^T \varepsilon$

which is a positive definite function. Hence

$$\dot{\mathcal{L}} = \varepsilon^T \dot{\varepsilon}. \quad (22)$$

Substituting the value of $\dot{\varepsilon}$ from (20) in (22) one gets

$$\dot{\mathcal{L}} = \varepsilon^T [\dot{\zeta}_{av} + \zeta_{av} - \zeta - (L+I)\varepsilon]. \quad (23)$$

This is considered in the case of connected graphs. So $\varepsilon^T (L+I)\varepsilon \geq \lambda_{\min}(L+I) \|\varepsilon\|^2 = \|\varepsilon\|^2$, where $\lambda_{\min}(L+I)$ is the minimum eigenvalue of the matrix $(L+I)$, [15]. Hence

$$\dot{\mathcal{L}} \leq \varepsilon^T Y - \|\varepsilon\|^2 \quad (24)$$

where $Y = [\gamma, \dots, \gamma]^T$, $Y \in R^n$

and $[\varepsilon^T, \dots, \varepsilon^T]^T \leq \sqrt{n} \|\varepsilon\|$, $[\varepsilon^T, \dots, \varepsilon^T]^T \in R^{n \times n}$, which is Jensen's inequality. Using these conditions and rearranging (24) it gives

$$\dot{\mathcal{L}} \leq \sqrt{n} \|\varepsilon\| Y - \|\varepsilon\|^2 \quad (25)$$

Now for $\mathcal{L} \leq \sqrt{n} \|\varepsilon\| Y$ within the bounded set and using LaSalle invariance principle [11], it can be shown that $\dot{\mathcal{L}} < 0$. Hence the considered error is globally asymptotically stable.

C. Potential function based Control of AUVs

The inter AUV attractive or repulsive force is calculated by using the artificial potential function which is a simple but powerful method used for path following control of the flocking structure. Here each AUV is considered as a single and isolated particle, supposed to be emerged in a potential well. The potential field is generated from two sources which are as follows. One type of potential field is calculated among the AUVs themselves, and another is between the AUVs and desired trajectory points. This potential function is a function of the distance between the positions of AUVs, or the distance between the position of the AUV and the position of the desired path, or position of the AUV and the position of the obstacles. The combined potential function is a nonnegative function $U(\zeta)$. The distance r_{ij} between the i^{th} and the j^{th} AUV is

presented as $r_{ij} = \|\zeta_i - \zeta_j\|$. Hence, one of the properties of potential function is that the potential function $U(\zeta)$ is closely related to the algebraic distance r_{ij} . The attractive or repulsive forces can be calculated from the corresponding attractive or repulsive potential functions respectively. These potential forces keep the multi AUV system to remain connected in the group. Also these forces will able the AUVs to travel along the desired path.

The magnitude of the repulsive force depends upon the magnitude of r_{ij} . The value of the repulsive force increases as the distance between the i^{th} and the j^{th} AVUs i.e. r_{ij} decreases to avoid collision. Repulsive force grows in unbounded manner to infinity as r_{ij} decreases and reaches zero value.

To avoid group splitting, an attractive potential function should exist between the flockmates. This cohesive force is an inter AUV distance dependent function. The value of this force should be zero when $\zeta_i = \zeta_j$ (no attractive force) and possess maximum value when $\|\zeta_i - \zeta_j\| \geq d$ (highest attractive force).

d is the safe distance. Here, all the AUVs of the group navigate towards a common flocking center. Hence, this potential function is a function of the positions of AUVs as well as their positions of the flocking centers. For the i^{th} AUV, it may be presented as $U_{i,att}^c(\zeta_i, \zeta_i^c)$.

Consider the attractive potential function for the i^{th} AUV as

$$U_{i,att}^c(\zeta_i, \zeta_i^c) = \frac{1}{2} k_{att}^c r_{ic}^2 \quad (26)$$

where $r_{ic} = \|\zeta_i - \zeta_i^c\|$ is the Euclidean norm between the position of the i^{th} AUV and the flocking center corresponding to the i^{th} AUV. Let k_{att}^c is a positive scaling factor. The gradient of this function can be given as

$$\nabla U_{i,att}^c(\zeta_i, \zeta_i^c) = k_{att}^c (\zeta_i - \zeta_i^c). \quad (27)$$

The attractive force between the position of the i^{th} AUV and position of corresponding flocking center is the negative gradient of the attractive potential function between them and can be given as [12], [17], [29],

$$F_{i,att}^c = -\nabla U_{i,att}^c(\zeta_i, \zeta_i^c) = -k_{att}^c (\zeta_i - \zeta_i^c). \quad (28)$$

Similarly an attractive potential function between the i^{th} AUV and the desired trajectory is taken as

$$U_{i,att}^d(\zeta_i, \zeta_d) = \frac{1}{2} k_{att}^d r_{id}^2 \quad (29)$$

where $r_{id} = \|\zeta_i - \zeta_d\|$ is the Euclidean distance between position of the i^{th} AUV and the desired trajectory. k_{att}^d is a positive scaling factor.

The potential force $F_{i,att}^d$ between the position ζ_i of the i^{th} AUV and the position vector of the desired trajectory ζ_d is

$$F_{i,att}^d = -\nabla U_{i,att}^d(\zeta_i, \zeta_d) = -k_{att}^d (\zeta_i - \zeta_d). \quad (30)$$

To avoid collision among the AUVs, a separation potential function is introduced within the group. The negative gradient

of this potential function is the separation potential force. When there is a chance of collision between AUVs i.e. when $\zeta_i = \zeta_j$, this potential force possesses maximum value. The value of the force asymptotically converges to zero when $\|\zeta_i - \zeta_j\| \geq d$. Hence AUVs would be able to keep a certain safe distance between them. The repulsive potential function between the i^{th} and the j^{th} AUV can be chosen as $U_{rep}^a(\zeta_i, \zeta_j)$. This function for the i^{th} AUV due to the j^{th} AUV in the group is $\sum_{j=1}^{n_i} U_{i,rep}^a(\zeta_i, \zeta_j) i \neq j$. If any AUV (either leader or follower) is at a larger distance than the safe distance from another AUV, the first AUV will be unable to repel the second one i.e. the repulsive potential is zero. The value of the repulsive potential function increases gradually when the AUVs move towards each other and diminishes gradually when they start apart from each other. At a certain distance the repulsive potential becomes zero. The repulsive potential force generated by the i^{th} AUV $F_{i,rep}^a$ due to their neighbor AUVs is the negative gradient of the repulsive potential function which is given by

$$F_{i,rep}^a = - \sum_{j=1}^{n_i} \nabla U_{i,rep}^a(\zeta_i, \zeta_j) \quad i \neq j. \quad (31)$$

In an obstacle-rich environment for collision avoidance, repulsive potential force between AUVs and the obstacles should be developed. In the similar way the negative gradient of potential function due to i^{th} AUV $F_{i,rep}^o$ can be found as

$$F_{i,rep}^o = - \sum_{j=1}^{n_{obs}} \nabla U_{rep}^o(\zeta_i, \zeta_j^o) \quad (32)$$

where n_{obs} is the number of obstacles, ζ_j^o is the position of the j^{th} obstacle, $\zeta_j^o = [x_j^o, y_j^o, z_j^o, \phi_j^o, \theta_j^o, \psi_j^o]^T$, $U_{rep}^o(\zeta_i, \zeta_j^o)$ is the repulsive potential between the i^{th} AUV and the j^{th} obstacle.

The total potential function for the i^{th} AUV can be obtained by taking summation of both the attractive and repulsive potentials. For the i^{th} AUV, the total potential force is given by

$$F_i = F_{i,att}^c + F_{i,att}^d + F_{i,rep}^a + F_{i,rep}^o. \quad (33)$$

Substituting (28), (30), (31) and (32) in (33) one can get as,

$$F_i = -k_{att}^c(\zeta_i - \zeta_i^c) - k_{att}^d(\zeta_i - \zeta_d) - \sum_{j=1}^{n_i} \nabla U_{i,rep}^a(\zeta_i, \zeta_j) - \sum_{j=1}^{n_{obs}} \nabla U_{rep}^o(\zeta_i, \zeta_j^o). \quad (34)$$

The control law for the i^{th} leader AUV can be represented as

$$\dot{\zeta}_i^l = F_i^l + \dot{\zeta}_d. \quad (35)$$

The states are considered only for the i^{th} leader AUV in (35). Here

$$F_i^l = -k_{att}^c(\zeta_i^l - \zeta_i^{lc}) - k_{att}^d(\zeta_i^l - \zeta_d) - \sum_{j=1}^{n_i} \nabla U_{i,rep}^a(\zeta_i^l, \zeta_j^l) - \sum_{j=1}^{n_{obs}} \nabla U_{rep}^o(\zeta_i^l, \zeta_j^o) \quad (36)$$

where ζ_i^{lc} is the flocking center of the i^{th} leader AUV.

The follower AUVs do not have any information of the desired trajectory. Hence, the controller for the i^{th} follower AUV may be presented as

$$\dot{\zeta}_i^f = F_i^f \quad (37)$$

with

$$F_i^f = -k_{att}^c(\zeta_i^f - \zeta_i^{fc}) - k_{att}^d(\zeta_i^f - \zeta_d) - \sum_{j=1}^{n_i} \nabla U_{i,rep}^a(\zeta_i^f, \zeta_j^f) - \sum_{j=1}^{n_{obs}} \nabla U_{rep}^o(\zeta_i^f, \zeta_j^o) \quad (38)$$

where ζ_i^{fc} denotes the flocking center of the i^{th} follower AUV. The stability of the closed loop system is analyzed in Section V.

V. SEPARATION POTENTIAL FUNCTION

The term which is common to both (36) and (38) i.e.

$\sum_{j=1}^{n_i} \nabla U_{i,rep}^a(\zeta_i, \zeta_j)$ and $\sum_{j=1}^{n_{obs}} \nabla U_{rep}^o(\zeta_i, \zeta_j^o)$ denotes the separation potential function of the i^{th} leader or follower AUV. There are two different types of separation potential functions are considered; mathematical potential functions and fuzzy potential functions. Mathematical potential function method uses a particular predefined and calculated formula. From this formula the potential forces are determined. It is a deterministic method. However, in fuzzy potential function method, the potential functions as well as potential force (repulsive/attractive) are determined by exploiting fuzzy inferencing.

A. Mathematical Separation Potential Function

The repulsive potential function can be developed in different ways. We consider the following mathematical repulsive potential function for the i^{th} AUV as [13],

$$U_{i,rep}^a = \frac{1}{\left(\frac{\|r_{ij}\|}{d}\right)^2} + \log\left(\frac{\|r_{ij}\|}{d}\right)^2. \quad (39)$$

The negative gradient of x-coordinate of potential function described in (42) may be presented by

$$F_{i,rep}^{a,x} = -\nabla U_{i,rep}^a = -\frac{2(x_i - x_j)(r_{ij}^2 - d^2)}{r_{ij}^4} \quad (40)$$

where x_i and x_j are the x components of the i^{th} and the j^{th} AUVs respectively. r_{ij} is the distance between the i^{th} and the j^{th} AUVs. d is the desired safe distance taken between AUVs.

Similarly the repulsive potential function and its negative gradient between the i^{th} AUV and the obstacle may be given as

$$U_{i,rep}^{o,x} = \frac{1}{\left(\frac{\|r_{ij,obs}\|}{d}\right)^2} + \log\left(\frac{\|r_{ij,obs}\|}{d}\right)^2$$

and

$$F_{i,rep}^{o,x} = -\nabla U_{rep}^o = -\frac{2(x_i - x_{j_{obs}})(r_{ij_{obs}}^2 - d^2)}{r_{ij_{obs}}^4} \quad (41)$$

where $x_{j_{obs}}$ is the x component of the j^{th} obstacle, $r_{ij_{obs}}$ is the distance between the i^{th} AUV and the j^{th} obstacles.

B. Fuzzy Separation Potential Function

The meaning of fuzzy is the uncertain quantity. In this case, input and the output entities are considered as the blurred variables. As the used parameters are not the fixed quantities, the procedure of designing fuzzy controller requires restorative assumptions. Hence heuristic methods are introduced to design the controller which provides a formal methodology for presenting and implementing human's inquisitive knowledge in the form of controller for operating the plants.

Fuzzy rules are defined based on the initial information obtained from the simulation of mathematical potential functions with appropriate choice of universe discourse of variables used in the fuzzy controller design. Fuzzy control system is necessarily consists of the following three steps: (1) choosing the fuzzy input/output for the controller, (2) Performing the process and post-process (if required) which are essential for the controller inputs and outputs, (3) Designing of components of fuzzy controller. Out of all the steps, the third step i.e. design of components of fuzzy controller is the most essential and is explained below.

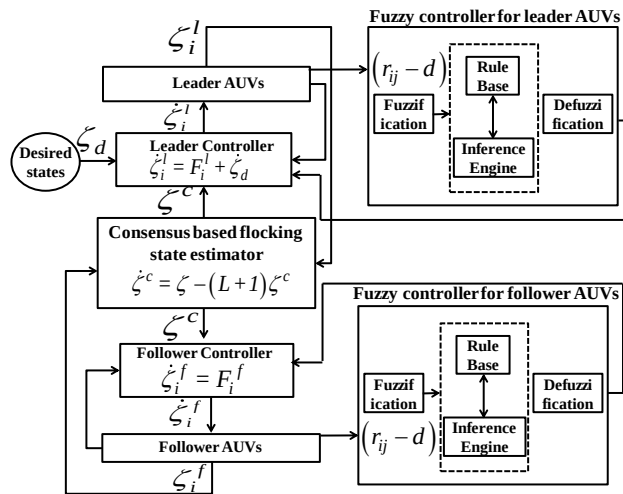


Fig. 3 Control structures for the flocking of AUVs

Generally, a fuzzy controller has four main components: (i) rule base, (ii) inference system, (iii) fuzzification, and (iv) defuzzification. The functions of these components are as follows. (i) Rule base: The rule base holds the knowledge of the available quantities in the form of a set of rules. This is the human approximation method which boosts the system to produce the best output and hence formulate the system to be the best. This is the quantification of the expert's thinking or expert's linguistic description and arrangement of this quantities in the form of a set of rules. (ii) Inference system: In this step, out of a set of rules most appropriate rule is selected for the current situation. This emulates the designer's thinking to make the controller performance the best. (iii) Fuzzification:

In this step, the crisp input is converted to the blurred information so that, the inference system can easily understand and activate the rules. These inputs are modified so that these can be interpreted and compared with the rule base. (iv) Defuzzification: In this step, the conclusions of the inference system which are in blurred form are converted to the crisp and solid outputs. These outputs act as the inputs for the plant.

Fuzzy control structure for the flocking of AUVs is shown in the Fig. 3. Mamdani's fuzzy approach is used for finding out the crisp output. The repulsive potential function between the i^{th} and the j^{th} AUVs is chosen as $U_{rep}^a(\zeta_i, \zeta_j)$. It is considered as a function of the input of the fuzzy system $(r_{ij} - d)$. The input crisp values, i.e. the distance error $(r_{ij} - d)$ are fuzzified and transformed to fuzzy sets. This input is converted into the respective degree of fuzzy membership functions corresponding to linguistic variables in fuzzy sets. These membership functions are the basic facts of the fuzzy inference process. Here the triangular membership functions are used because of its simplicity and easy computation method. The crisp input function is divided into five sets of fuzzy linguistic variables and the triangular membership functions are chosen as shown in Fig. 4 (a). Similarly the output is the potential function between i^{th} and j^{th} AUVs as shown in Fig. 4 (b). In Fig. 4 (a) and 4 (b); NB, N, Z, P, PB are the abbreviations of negative big, negative, zero, positive and positive big respectively.

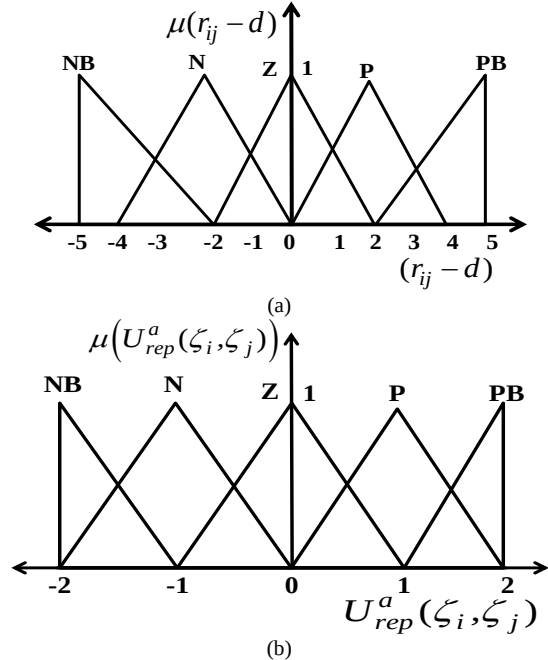


Fig. 4. (a) Input fuzzy variable with triangular membership function, (b) Output fuzzy variable with triangular membership function.

The relationship between the distance error (in terms of safe distance i.e. $(r_{ij} - d)$) and the output, i.e. artificial potential functions is related by a set of rule base which is known as fuzzy IF-THEN rule base whose general format is presented as

R_i : IF H_i is h_i , THEN G_i is g_i

where $h_1, h_2, \dots, h_k$ are the fuzzy sets corresponding to the

input linguistic variables $H_1, H_2, \dots, H_k$ respectively. Hence $h_i \in \{NB, N, Z, P, PB\}$ and $r_{ij} \in \{H_1, H_2, \dots, H_k\}$. Similarly g_i are the fuzzy sets applied to the output linguistic variables G_i of rule R_i , so $g_i \in \{NB, N, Z, P, PB\}$ and $U_{rep}^a(\zeta_i, \zeta_j) \in \{G_1, G_2, \dots, G_k\}$. The method of fuzzy rule evaluation may be explained as follows. The degree of every predicate which are the elements of rule R_i are explained as:

The degree of membership of ' H_1 in h_1 ' is m_1
 The degree of membership of ' H_2 in h_2 ' is m_2
 $\vdots$
 The degree of membership of ' H_k in h_k ' is m_k

The overall degree of the membership of the premises can be calculated by taking the minimum of the membership of all the individual predicates present in the premises of the rule R_i . This is the Mamdani's rule of the fuzzy set theory. Hence the overall degree of membership m_i of the premises of the rule R_i is

$$m_i = \text{Min}\{m_1, m_2, \dots, m_k\} \quad (42)$$

In this particular case, the distance between the i^{th} and the j^{th} AUVs is considered as r_{ij} and the safe distance between them is d . If $r_{ij} > d$ then there is a chance of breaking of the group. Hence, for keeping the group unbroken, the potential force (negative gradient of the potential function) between the AUVs is attractive and this is reversed for the case where $r_{ij} < d$. Referring to these conditions, the following rules can be formulated. If the input $(r_{ij} - d)$ is less, the output $U_{rep}^a(\zeta_i, \zeta_j)$ is less and if the input is more, the output is more i.e. the input is directly proportional to the output variable. In order to obtain matching of the input and output functions, a rule base is prepared which consists of a set of five rules which is given below.

Rule Base:

If input is NB then output is NB
 If input is N then output is N
 If input is Z then output is Z
 If input is P then output is P
 If input is PB then output is PB

Then each output of the fuzzy system is defuzzified to give the deterministic crisp value which is the most expected output of the fuzzy system. This step converts a set of modified control output value into a single point wise crisp value. Here the center of gravity (COG) method is implemented to obtain the single deterministic crisp value from the bunch of probabilistic fuzzified values. With this method the crisp value β_p for the system Y_p can be computed as follows.

Let us consider $Q_p = [a, b]$ as the universe of discourse and the linguistic variable is Y_p for the p^{th} fuzzy subsystem. For this linguistic variable a and b are the upper and lower limits respectively. Hence ' Y_p is β_p ' is a composite conclusion within the range $[a, b]$. The crisp deterministic value of the Y_p is the amalgamation of all possible defuzzified subsets within the range $[a, b]$. This crisp value can be calculated by implementing the center of gravity (COG) defuzzification

method which is presented as follows:

$$\beta_p = \frac{\int_a^b x \mu(x) dx}{\int_a^b \mu(x) dx} \quad (43)$$

where $\mu(x)$ is the degree of membership of the function x . In this case the function x is $x = (r_{ij} - d)$ for input or $x = U_{rep}^a(\zeta_i, \zeta_j)$ for output.

The flowchart of the proposed flocking control algorithm is presented in Fig. 5.

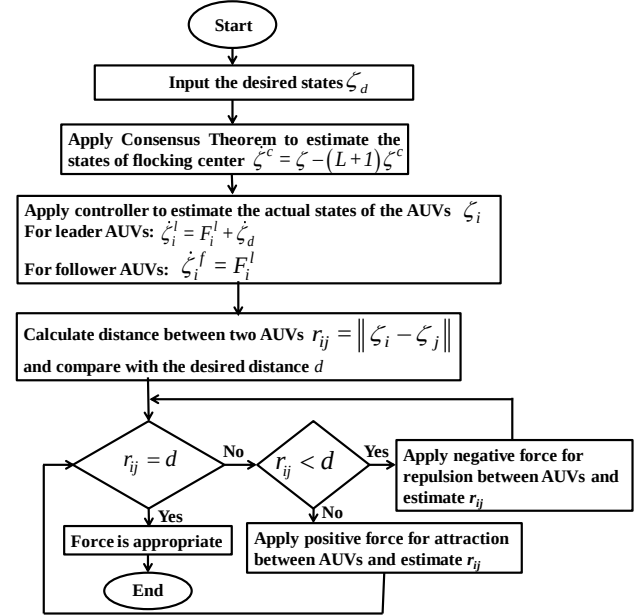


Fig. 5. Flowchart of flocking algorithm

With the LaSalle's invariance principle the developed controllers in (36) and (38) are ensured to be stable.

Proof:

The attractive potential function due to a single AUV is given in (26). So the potential function due to n AUVs may be given as

$$U_{att}^c = \sum_{i=1}^n U_{i,att}^c = \sum_{i=1}^n \frac{1}{2} k_{att}^c r_{ic}^2 \quad (44)$$

As $r_{ic} = \|\zeta_i - \zeta_i^c\|$, the first derivative of this function (44) may be given by

$$\dot{U}_{att}^c = \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T \dot{\zeta}_i - \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T \dot{\zeta}_i^c \quad (45)$$

Similarly the time derivative of attractive potential function between leader AUVs and the position coordinates of the desired trajectories is obtained from (29) as

$$U_{att}^d = \sum_{i=1}^n U_{i,att}^d = \sum_{i=1}^n \frac{1}{2} k_{att}^d r_{id}^2 \quad (46)$$

Hence

$$\dot{U}_{att}^d = \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_i^l - \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d. \quad (47)$$

The time derivative of repulsive potential function among the AUVs can be obtained from (31) as

$$\dot{U}_{rep}^a = \sum_{i=1}^n \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i} (U_{rep}^a (\zeta_i, \zeta_j))^T \dot{\zeta}_i. \quad (48)$$

The repulsive potential among the AUVs and the obstacles is

$$\dot{U}_{rep}^o = \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i} (U_{rep}^o (\zeta_i, \zeta_j^o))^T \dot{\zeta}_i. \quad (49)$$

The mutual attractive potential function between the i^{th} and the j^{th} AUVs can be written as

$$\nabla_{\zeta_i} (U_{rep}^a (\zeta_i, \zeta_j))^T \dot{\zeta}_i = -\nabla_{\zeta_j} (U_{rep}^a (\zeta_i, \zeta_j))^T \dot{\zeta}_j. \quad (50)$$

Thus, the derivative of the total potential function is

$$\dot{U} = \dot{U}_{att}^c + \dot{U}_{att}^d + \dot{U}_{rep}^a + \dot{U}_{rep}^o. \quad (51)$$

Substituting the value of (45), (47) (48) and (49) in (51), one obtains

$$\begin{aligned} \dot{U} = & \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T \dot{\zeta}_i - \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T \dot{\zeta}_i^c \\ & + \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_i^l - \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i} (U_{rep}^a (\zeta_i, \zeta_j))^T \dot{\zeta}_i + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i} (U_{rep}^o (\zeta_i, \zeta_j^o))^T \dot{\zeta}_i \end{aligned} \quad (52)$$

Distinguishing the group state vector (ζ_i) , leader state vector (ζ_i^l) and follower state vector (ζ_i^f) , (52) can be rewritten as

$$\begin{aligned} \dot{U} = & \sum_{i=1}^{n_l} k_{att}^c (\zeta_i^l - \zeta_i^c)^T \dot{\zeta}_i^l - \sum_{i=1}^{n_l} k_{att}^c (\zeta_i^l - \zeta_i^c)^T \dot{\zeta}_i^c \\ & - \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T \dot{\zeta}_i^c + \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_i^l \\ & - \sum_{i=1}^{n_f} k_{att}^c (\zeta_i^f - \zeta_d)^T \dot{\zeta}_d + \sum_{i=1}^{n_l} \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^l} U_{rep}^a (\zeta_i^l, \zeta_j) \dot{\zeta}_i^l \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^l} (U_{rep}^o (\zeta_i^l, \zeta_j^o))^T \dot{\zeta}_i^l + \sum_{i=1}^{n_f} \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^f} U_{rep}^a (\zeta_i^f, \zeta_j) \dot{\zeta}_i^f \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^f} (U_{rep}^o (\zeta_i^f, \zeta_j^o))^T \dot{\zeta}_i^f \end{aligned} \quad (53)$$

Substituting (11), (34), (35), (36) and (50) in (53), one can get as;

$$\begin{aligned} \dot{U} = & - \sum_{i=1}^{n_l} \left\| k_{att}^c (\zeta_i^l - \zeta_i^{lc})^T + k_{att}^d (\zeta_i^l - \zeta_d)^T \right\|^2 \\ & + \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^l} (U_{rep}^a (\zeta_i^l, \zeta_j)) \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^l} (U_{rep}^o (\zeta_i^l, \zeta_j^o))^T \dot{\zeta}_i^l \\ & - \sum_{i=1}^{n_f} \left\| k_{att}^c (\zeta_i^f - \zeta_i^{fc})^T + \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^f} (U_{rep}^a (\zeta_i^f, \zeta_j)) \right\|^2 \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^f} (U_{rep}^o (\zeta_i^f, \zeta_j^o))^T \dot{\zeta}_i^f \\ & - \sum_{i=1}^n \|\zeta_i - \zeta_i^c\|^2 + \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T (\zeta_i^c - \zeta_i^c)^T \\ & + \sum_{i=1}^{n_l} k_{att}^c (\zeta_i^l - \zeta_i^{lc})^T \dot{\zeta}_d + \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d \\ & + \sum_{i=1}^{n_l} \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^l} (U_{rep}^a (\zeta_i^l, \zeta_j)) \dot{\zeta}_d - \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d \end{aligned} \quad (54)$$

Eq. (53) may be presented by

$$\begin{aligned} \dot{U} = & -\Lambda + \sum_{i=1}^{n_l} k_{att}^c (\zeta_i^l - \zeta_i^{lc})^T \dot{\zeta}_d + \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d \\ & + \sum_{i=1}^{n_l} \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^l} (U_{rep}^a (\zeta_i^l, \zeta_j)) \dot{\zeta}_d \\ & + \sum_{i=1}^n k_{att}^c (\zeta_i - \zeta_i^c)^T (\zeta_i^c - \zeta_i^c)^T - \sum_{i=1}^{n_l} k_{att}^d (\zeta_i^l - \zeta_d)^T \dot{\zeta}_d \end{aligned} \quad (55)$$

where

$$\begin{aligned} \Lambda = & \sum_{i=1}^{n_l} \left\| k_{att}^c (\zeta_i^l - \zeta_i^{lc})^T + k_{att}^d (\zeta_i^l - \zeta_d)^T \right\|^2 \\ & + \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^l} (U_{rep}^a (\zeta_i^l, \zeta_j)) \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^l} (U_{rep}^o (\zeta_i^l, \zeta_j^o))^T \dot{\zeta}_i^l \\ & + \sum_{i=1}^{n_f} \left\| k_{att}^c (\zeta_i^f - \zeta_i^{fc})^T + \sum_{j=1}^{n_{ni}} \nabla_{\zeta_i^f} (U_{rep}^a (\zeta_i^f, \zeta_j)) \right\|^2 \\ & + \sum_{i=1}^n \sum_{j=1}^{n_{obs}} \nabla_{\zeta_i^f} (U_{rep}^o (\zeta_i^f, \zeta_j^o))^T \dot{\zeta}_i^f + \sum_{i=1}^n \|\zeta_i - \zeta_i^c\|^2 \end{aligned} \quad (56)$$

Analyzing (55) and (56) it is observed that $\dot{U} \leq -\Lambda$. From these results it is clear that by LaSalle's invariance principle, $\dot{U}$ decreases gradually until $\dot{U} = 0$. This expression ensured the stability conditions of the developed controller.

VI. RESULTS AND DISCUSSION

A simulation set up is made for studying the efficacy of the proposed leader-follower flocking algorithm. Different cases are considered for flocking control of four AUVs. In one case, all the four AUVs are considered as leader vehicles, i.e., each AUV has global knowledge of the desired path. In another case, out of four AUVs in a group, only one AUV is considered as the leader vehicle and other AUVs are considered as followers. The simulation is carried out for both the mathematical and fuzzy separation potential function methods and the results are compared. In both the cases the desired trajectory is defined as a circular path of radius 10m and in a space of height 10m which is given by;

$$x_d = 10\sin(0.01t) \quad y_d = 10\cos(0.01t) \quad z_d = 10 \quad \psi_d = \frac{\pi}{3}. \quad (57)$$

The constants are taken as $k_{att}^c = k_{att}^d = 10$, $d = 2m$.

The pitch and roll orientations are neglected here as these parameters are not much imperative and simulations are carried out considering four DOFs.

In case of obstacle avoidance, the obstacles are taken as solid spherical obstacles having radius 1.5m.

A. Mathematical Separation Potential Function based flocking control

Two cases such as; all the four AUVs are considered as leaders and one AUV is taken as leader with other three AUVs are considered as followers.

Case 1(a): Flocking of Four AUVs (All are leaders) in space using the mathematical separation potential function in Circular path in space.

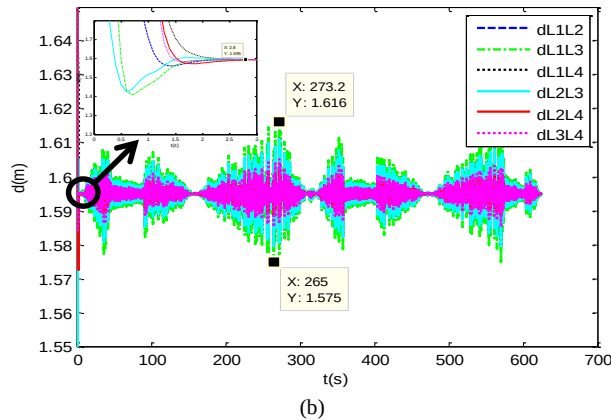
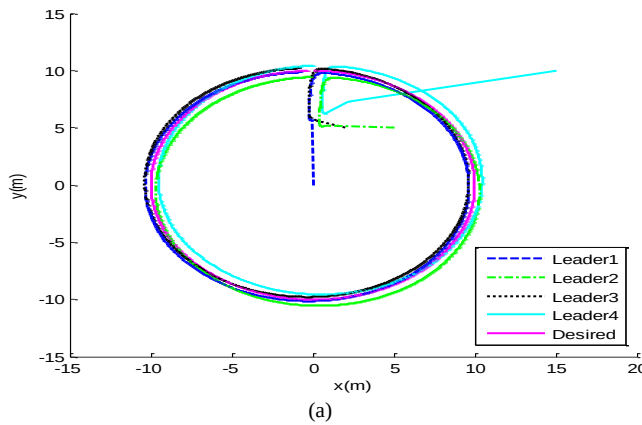


Fig. 6. (a) Flocking of four AUVs (all are leaders) is travelling on the circular path using a separation potential function (b) Distance among AUVs in 650s

Case 1(b): Flocking of Four AUVs (All are leaders) in space using the mathematical separation potential function in Circular path in space with 10% external oceanic disturbances.

The control law for the i^{th} leader AUV with 10% external oceanic disturbances can be represented as

$$\dot{\zeta}_i^l = F_i^l + F_i^{ld} + \dot{\zeta}_d^l. \quad (58)$$

where F_i^{ld} presents the external oceanic disturbance and it is considered as 10% of F_i^l .

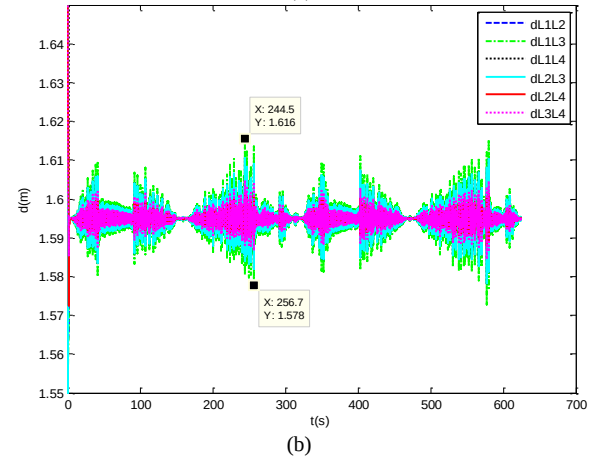
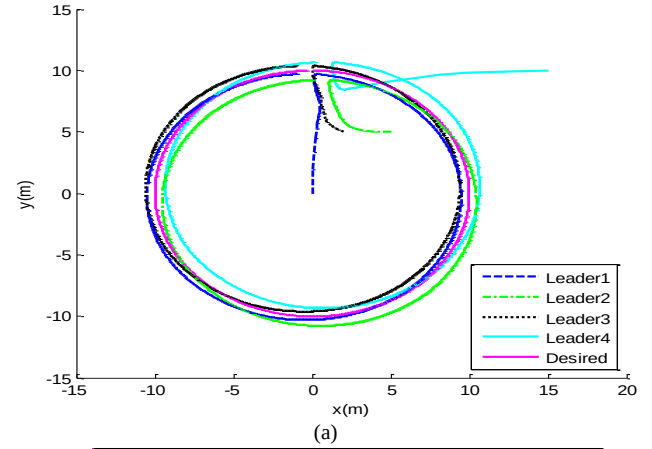


Fig. 7. In 10% external oceanic disturbances (a) Flocking of four AUVs (all are leaders) (b) Distance among AUVs in 650s

Fig. 7 presents the simulation results of the flocking of the four AUVs in presence of 10% external oceanic disturbances. From the figures 6 and 7 it is observed that, there is small variation in the results between the presence and absence of external oceanic disturbances.

Case 2: Flocking of four AUVs (one leader and three followers) in a circular path in space

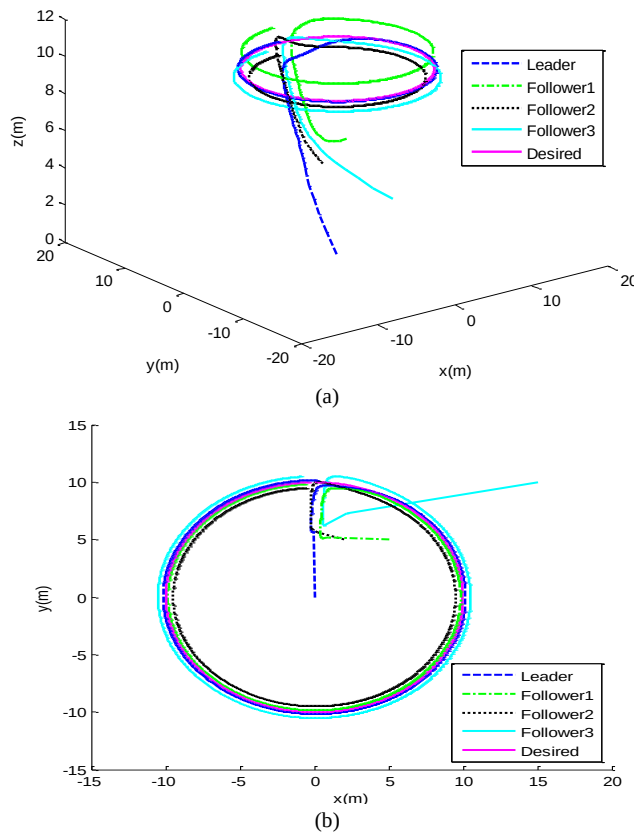


Fig. 8. Flocking of AUVs (one leader and three followers) on the circular path in (a) space (b) plane

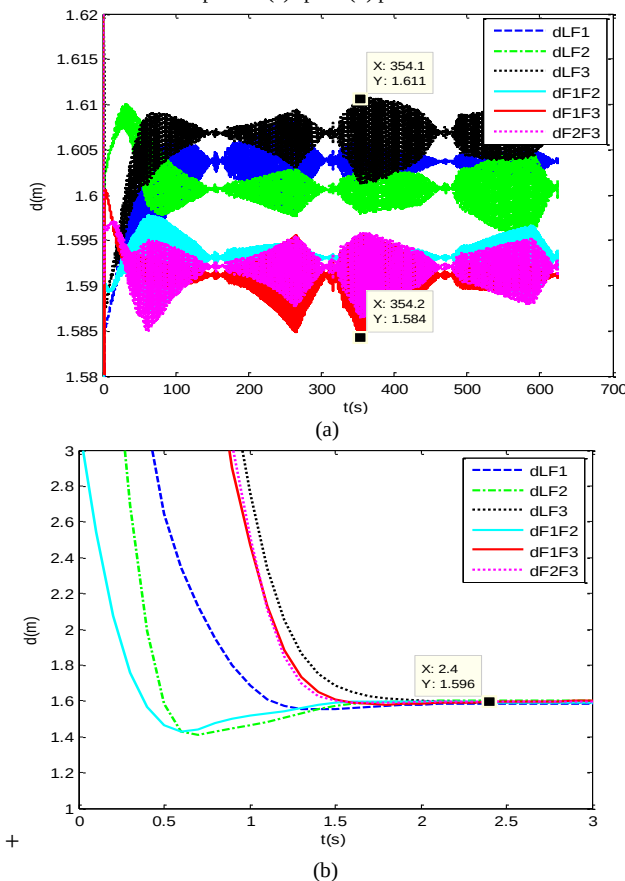


Fig. 9. Distance among the AUVs on the circular for (a) 650s. (b) 3s.

Fig. 8 shows flocking of AUVs along the desired track and Fig. 9 presents the distances maintained among the AUVs.

Fig. 9 shows distance among four AUVs for the case 2 in (a) 650s and (b) 3s. From Fig.8 and Fig.9, it is observed that all the four AUVs are able to track the desired trajectory obeying the leader-follower flocking law without facing any collision among them. The followers are able to follow the leader to track the desired trajectory.

B. Fuzzy Separation Potential Function based flocking control

As presented in previous part of this section, there are two cases considered for leader-follower flocking control using fuzzy separation potential functions. The range of the universe of discourse for input is $[-5, 5]$ and that of the output is $[-2, 2]$.

Case 1(a): Flocking of four AUVs in a desired circular path with all Four AUVs are considered as leaders.

Fig. 10 (a) and Fig. 10 (b) show the tracking of the desired circular path by a flock of four AUVs in three dimensions and in two dimensions respectively. In both the cases all the four AUVs are considered as leaders, i.e. all AUVs have knowledge of the desired trajectory. From these figures it is observed that the group of AUVs are flocking as a whole along the desired circular path without facing any collision among them.

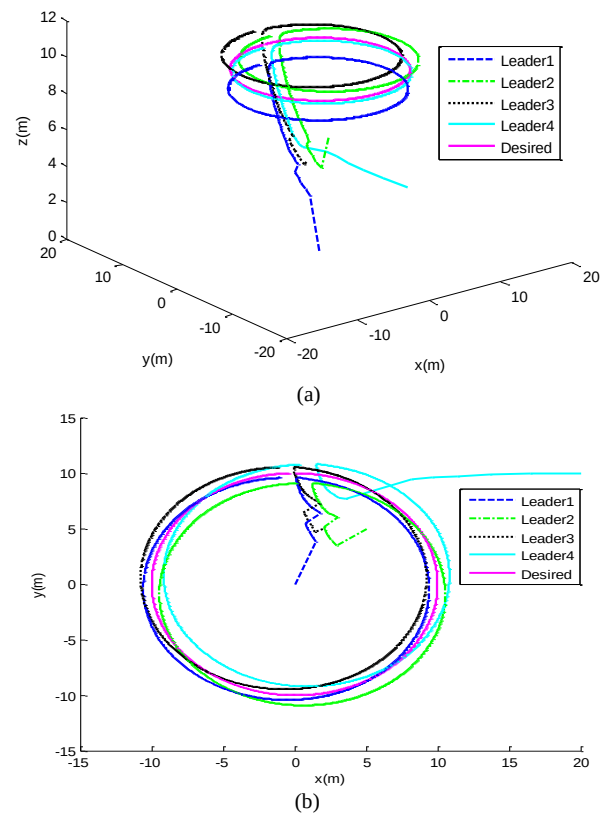


Fig. 10. Flocking of AUVs considering all are leaders on the circular path in (a) space (b) plane.

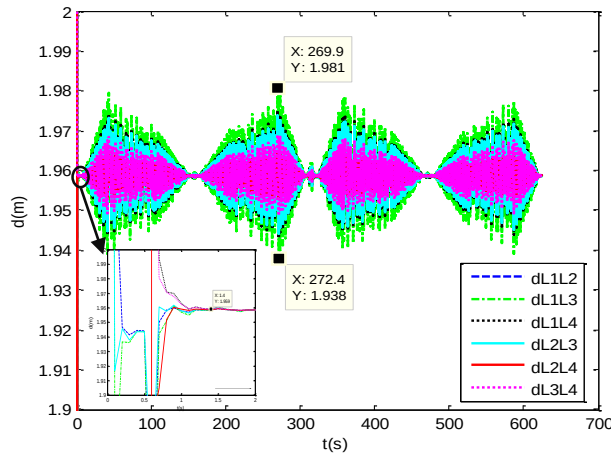


Fig. 11. Distance among AUVs for case 1

From Fig. 11, it is observed that during flocking along the desired path, AUVs separate from each other and avoid collision among themselves. Here dL_iL_j denotes the distance between the i^{th} and the j^{th} leader AUVs, $1 \leq i \leq 4, 1 \leq j \leq 4$.

Case 1(b): Flocking of four AUVs in a desired circular path with all Four AUVs are considered as leaders with 10% external environmental disturbances.

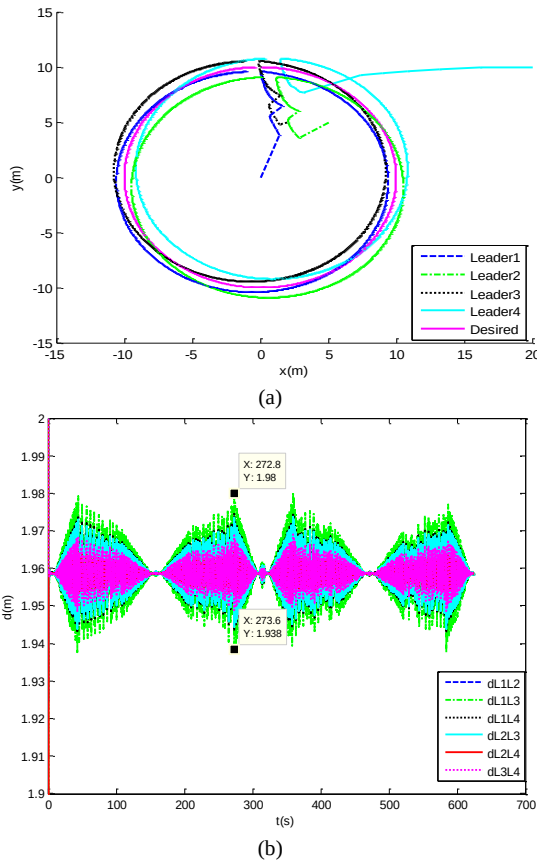


Fig. 12. In 10% external oceanic disturbances (a) Flocking of four AUVs (all are leaders) is travelling on the circular path using a separation potential function (b) Distance among AUVs in 650s

In presence of 10% external oceanic disturbances by using the fuzzy potential functions, the simulation results of the flocking of the four AUVs are presented in Fig. 12. By analyzing the figures 10, 11 and 12 it is observed that, there is

very little variation in the results between the presence and absence of external oceanic disturbances, which may be considered as no variation. Hence it is verified that when there is a deviation in uncertainty due to external disturbances such as varying hydrodynamics damping and ocean currents, the proposed flocking controller provides good performance despite these variations in external disturbances.

Case 2: Flocking of four AUVs along the desired circular path with one leader AUV and three follower AUVs in a group.

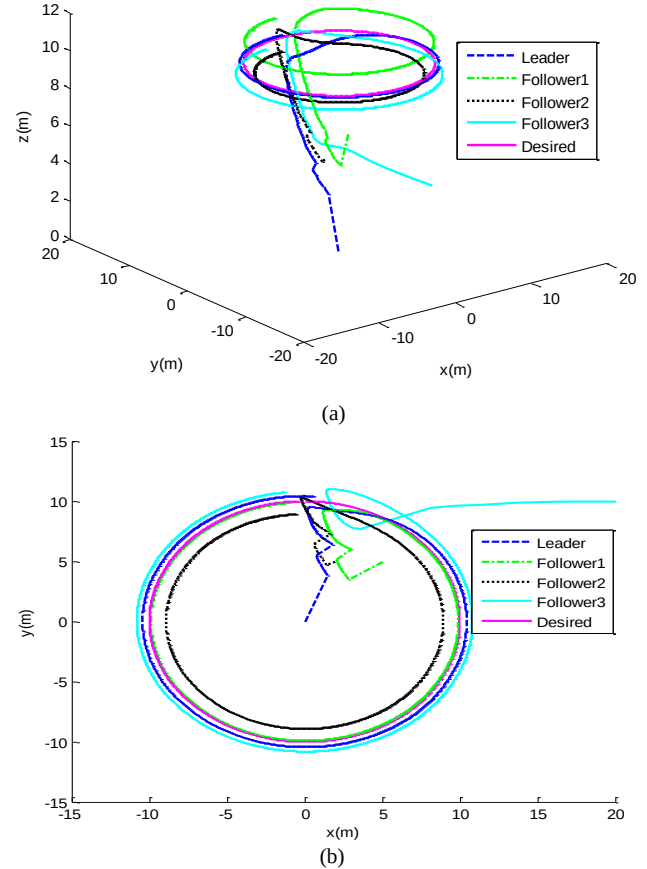


Fig. 13. Flocking of AUVs considering one leader AUV and three follower AUVs in (a) space (b) plane.

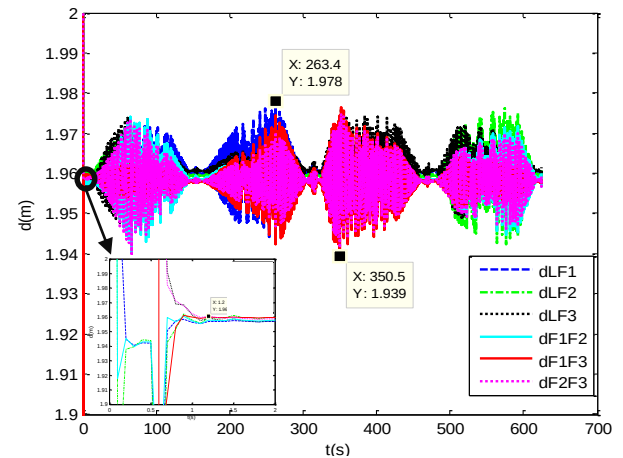


Fig. 14. Distance among AUVs for case 2

Fig. 13 (a) shows the three dimensional trajectory tracking of

group of four AUVs flocking along the desired circular path in which there are three AUVs are considered as followers and one AUV is considered as leader. It is observed from this figure that, this group of AUVs starts from arbitrary points and successfully tracks the desired trajectory which is given in (60). Fig. 13 (b) shows the group consisting of one leader and three follower AUVs flocking along the desired circular path in the plane. Fig. 14 shows the distances maintained between leader and follower AUVs. In these figures, dLF_i represents the distance between the leader and the i^{th} follower AUV, and $dFiFj$ is the distance between the i^{th} and the j^{th} follower AUVs, $1 \leq i \leq 3$. From these figures it is observed that the group of AUVs flocks along the desired path without any collision.

C. : Obstacle avoidance of four AUVs (One leader and three followers) using mathematical functions

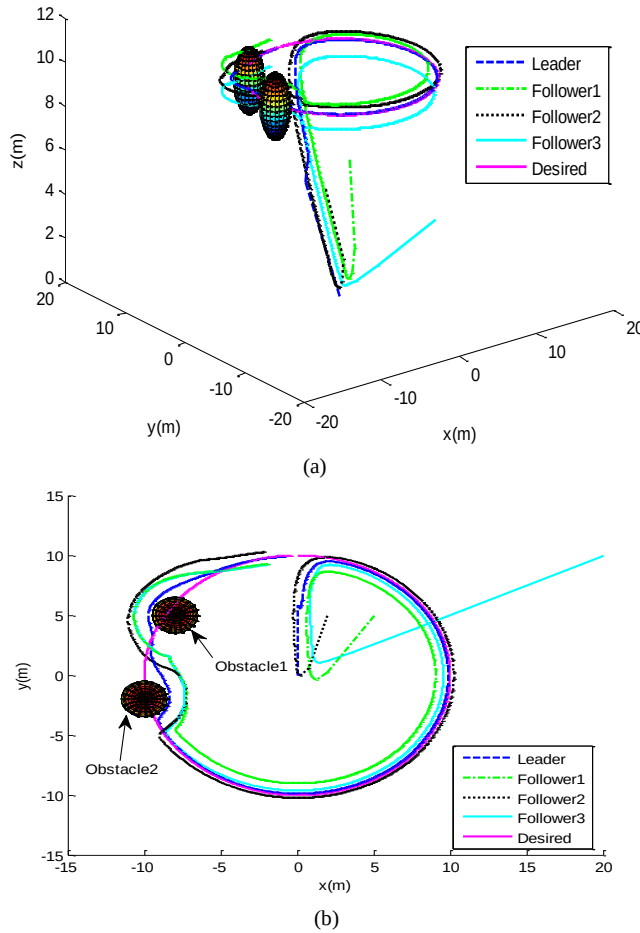


Fig. 15. Flocking of four AUVs (one leader and three followers) travelling on the circular path and avoiding solid obstacles in obstacle-rich environment in (a) space (b) plane using mathematical potential functions

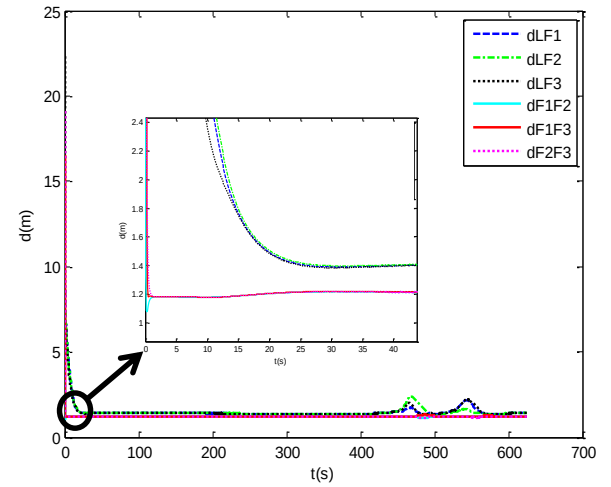


Fig. 16. Distance among four AUVs (one leader and three followers) during Flocking on the circular path during obstacle avoidance using mathematical potential functions.

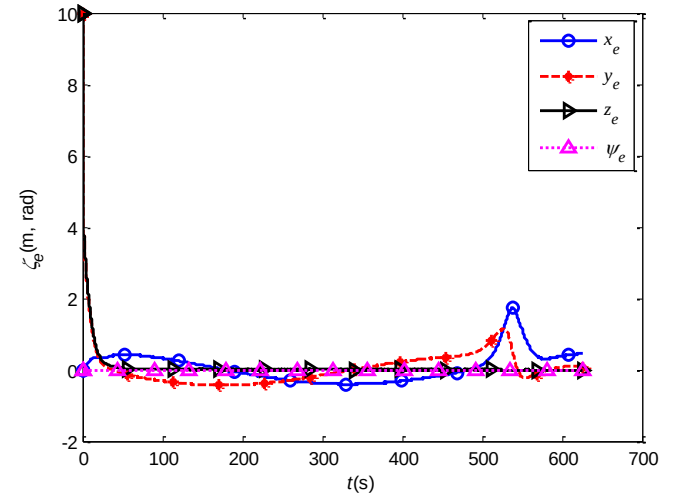
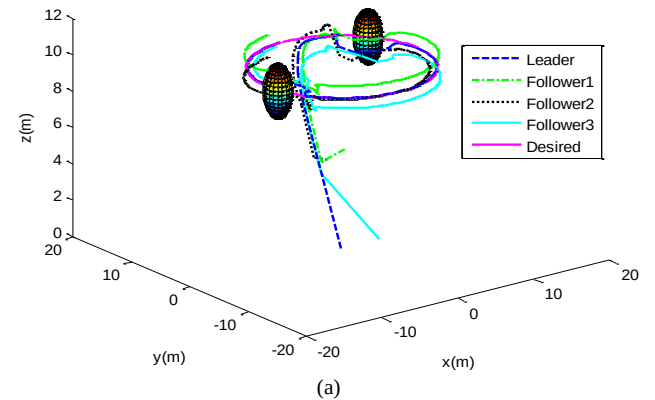


Fig. 17. Position errors during Flocking of four AUVs using mathematical potential functions.

D. : Obstacle avoidance of four AUVs (One leader and three followers) using fuzzy functions



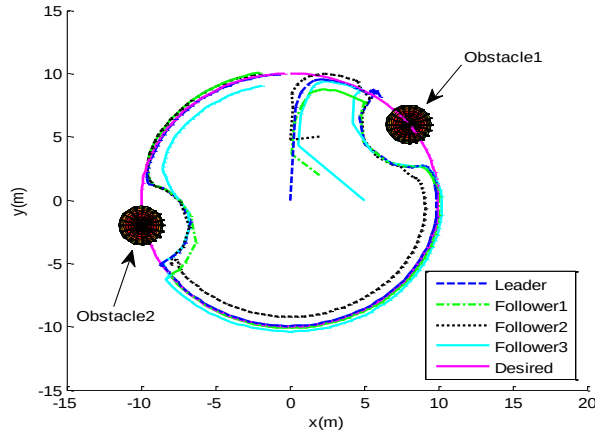


Fig. 18. Flocking of four AUVs (one leader and three followers) travelling on the circular path and avoiding solid obstacles in (a) space (b) plane using fuzzy potential functions

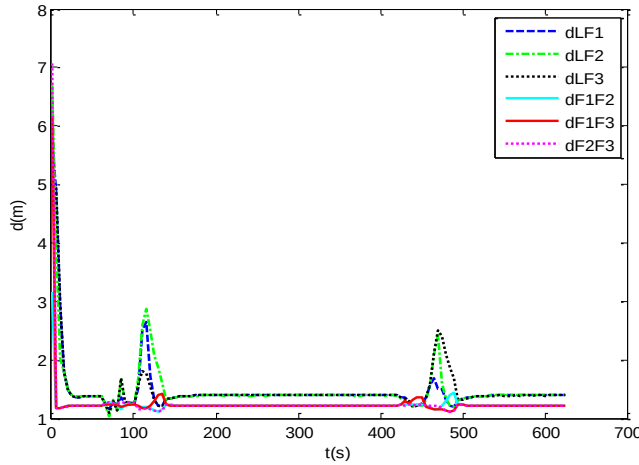


Fig.19. Distance among four AUVs (one leader and three followers) during Flocking on the circular path with obstacle avoidance using fuzzy potential functions.

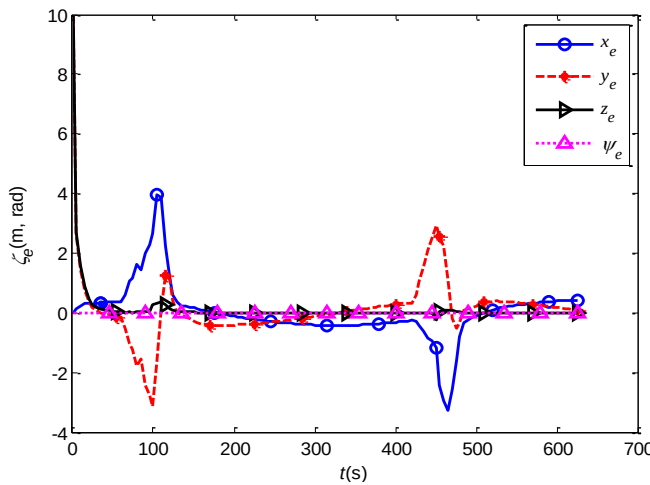


Fig. 20. Position errors during Flocking of four AUVs using fuzzy potential functions.

Fig. 15 and Fig. 16 present the simulation results of the flock of AUVs in an obstacle-rich region using mathematical and fuzzy separation potential functions respectively. Here there are

one leader and three follower AUVs are considered. From these figures it is observed that the controller is able to operate the AUVs to avoid the obstacles safely. The distance among the AUVs during flocking in the obstacle-rich environment is provided in Fig. 16 and Fig. 19 corresponding to mathematical and fuzzy separation potential functions respectively. Fig. 17 and Fig. 20 present the position errors present during flocking of AUVs using mathematical and fuzzy controllers respectively. From these figures it is observed that, there is no chance of collision among AUVs or AUV with obstacles arisen during flocking.

Comparison:

From Figures 6(a), 8, 10 and 12, it is observed that during tracking along the desired trajectories, AUVs operate by the flocking control laws based upon the mathematical and fuzzy separation potential functions. From the figures 6(b), 9, 11 and 14, the observed flocking approach times and the distance among AUVs are mentioned in the Table 1.

During flocking, two important considerations are made: (i) Flocking approach time (ii) Safe distance. (i) Flocking approach time: The time taken by the AUVs to reach on the desired trajectory starting from an initial point. The flocking approach time should be less for better performance of the controller and vice versa. (ii) Safe distance: The distances among the AUVs should be maintained close to the safe distance so that neither the collision among the AUVs nor the collision of the AUVs with the solid obstacles would happen. Apart from this, by maintaining the safe distance among the AUVs, the chances of group splitting could be avoided. Hence, more the error in maintaining the safe distance, more the chances of collision and group splitting and vice versa.

TABLE 1. COMPARISONS OF FLOCKING APPROACH TIMES AND DISTANCES AMONG AUVs FOR BOTH THE METHODS

Control strategy for different cases	Mathematical function based Flocking			Fuzzy function based Flocking		
	Flocking approach time (Sec)	Distance between AUVs (m)		Flocking approach time (Sec)	Distance between AUVs (m)	
		Minimum	Maximum		Minimum	Maximum
All four are leaders	2.9	1.575	1.616	1.4	1.938	1.981
One leader and three followers	2.4	1.584	1.611	1.2	1.939	1.978

From Table 1 it is observed that during flocking, the flocking approach time in the case of mathematical method is more than that of fuzzy method. From this Table 1 it is also observed that, the distances sustained among the AUVs due to application of fuzzy potential function is closer to the desired safe distance. Obviously it is clear that the flocking controller developed by using fuzzy potential functions outperforms the flocking controller developed using mathematical potential functions.

VII. CONCLUSIONS

A leader follower algorithm for the flocking control of a

group of AUVs using both mathematical and fuzzy potential functions has been presented. A virtual point known as flocking center is calculated using consensus protocol. The entire system of AUVs remains bounded with the flocking center of the team. The repulsive potential between two AUVs is calculated using mathematical as well as fuzzy logic methods. In both the methods, potential functions are considered as the function of the inter AUV distances. The efficacy and accuracy of the proposed flocking algorithm are verified through simulations performed considering four AUVs in a group. The simulation is also carried out in an obstacle-rich environment. From the results obtained, it is observed that the AUVs flock in the desired path without facing any collision among them as well as with the obstacles. From the results as well as by comparing the performances, it is concluded that the fuzzy potential based flocking controller is more efficient than the mathematical potential function based flocking controller.

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